EUGENICS

Can Selective Reproduction Ever Be Scientifically or Ethically Justified?

A critical review of history, genetics, and bioethics

UNIVERSITY-LEVEL RESEARCH REPORT



Original writing and illustrations • Numbered citations • Full bibliography

Prepared 27 October 2026

Reader’s guide and central argument

> Answer in brief
> A scientifically meaningful reduction in inherited disease risk can sometimes support a voluntary reproductive choice. It does not establish the ethical legitimacy of coercion, population “improvement,” or a hierarchy of human worth. The justification must attach to a specific practice, not to eugenics as a blanket programme.

This report asks two separate questions: **what can genetics predict or change?** and **what may people or institutions permissibly do with that knowledge?** Evidence of heritability cannot, by itself, answer the second question. A defensible analysis must examine consent, disability, justice, clinical benefit, and the distribution of power. [6–9,18,24]

Structure

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Scope and source transparency

This is a critical narrative review, not a systematic review or individual medical advice. It uses peer-reviewed literature, historical scholarship, primary legal and historical records, and institutional guidance. The supplied epigenetics PDF informed presentation only; none of its subject content was reused.

Source access: Galton’s public-domain text, the full text of Lencz et al. in eLife’s official publication archive, and two author-maintained research repositories were directly inspected. [1,14,23,26] Most publisher and institutional websites returned network denials. Other works are cited from established literature with identifiers or links; their full texts and current web versions were not independently rechecked. This review is not exhaustive or a statement of current law in every jurisdiction.

A. Introduction and historical context

A.1 Definition and boundaries

Eugenics refers to efforts to shape the hereditary composition of a population through differential reproduction, historically justified as improving the population. Francis Galton coined the term in 1883. His account fused observations about inheritance with judgements about supposedly superior human "types," demonstrating that the project contained social values from its beginning. Those hierarchies are historical claims, not accepted scientific categories. [1,2,25]

Selection can change allele frequencies across generations: inheritance provides a real biological mechanism. It does not follow that complex social traits can be reliably engineered, or that a population has one genetic optimum. Predicting a response to selection and deciding whether that response is desirable are different tasks. [2,13–16,24]

"Selective reproduction" is broader: it includes decisions about whether, with whom, or using which gametes or embryos to reproduce. Not every use of genetic information has the same purpose or moral character. A couple's effort to avoid a particular serious disorder differs from a state programme that controls which groups may reproduce. Whether voluntary selection should itself be called eugenic remains disputed. [18,19,24]

Term	Meaning and necessary distinction
Historical "positive" eugenics	Encouraging reproduction among groups labelled desirable. "Positive" describes a method, not moral approval.
Historical "negative" eugenics	Reducing reproduction among people labelled undesirable, including through segregation and forced sterilization.
Voluntary reproductive genetics	Counselling, carrier testing, prenatal diagnosis, or embryo testing chosen without penalties or population targets.
Heritable genome editing	Changing DNA in gametes or embryos for reproduction, potentially passing the alteration to descendants. Selection itself does not rewrite DNA.

A.2 Why the distinction matters

The same laboratory test may sit within very different institutions. Counselling can expand choice; a mandate, insurance condition, marriage restriction, or stigmatizing public campaign can narrow it. Ethical assessment therefore needs both the technical details and the social setting. A label such as "voluntary" does not settle whether consent is meaningful. [7,11,18]

> Guiding distinction
> Reducing the probability of a serious condition is a possible medical goal. Ranking the worth of people, families, ancestries, or disabilities is a separate moral claim that genetic evidence cannot establish.

A.3 Historical development

Eugenics grew through interactions among hereditarian science, nationalism, public administration, and prejudice. It appeared in multiple countries and political systems. Its history cannot be reduced to Nazi Germany, nor can all national programmes be treated as identical. [2,3,5,25]

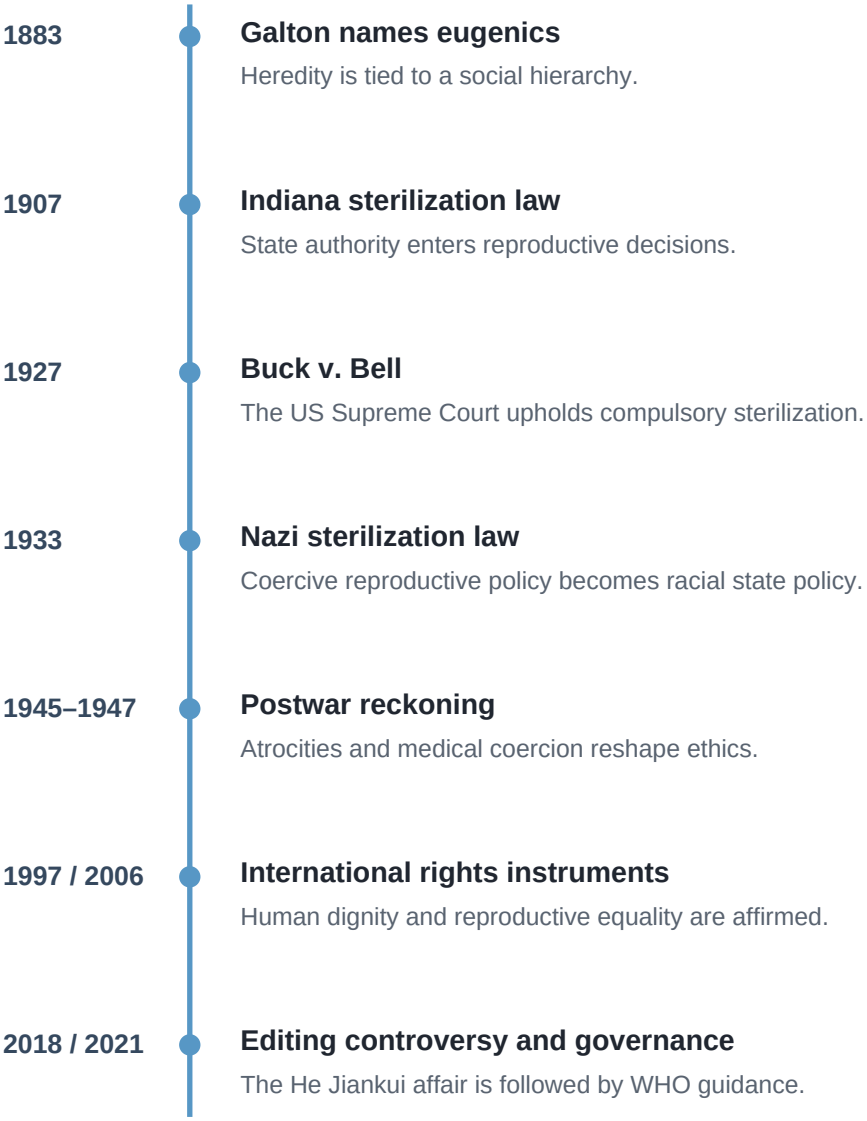


Figure 1. Selected milestones, not an exhaustive global chronology. Sources: Galton; Kevles; Stern; Buck v. Bell; USHMM; UNESCO; CRPD; Greely; WHO. [1–8,20]

A.4 What went wrong scientifically?

Many programmes treated poverty, criminality, institutionalization, and culturally defined behaviour as if they were simple inherited defects. Biased family records and classifications often confounded heredity with discrimination and living conditions. The history illustrates a recurring error: using a descriptive claim about inheritance to justify a social ranking and then a coercive policy. Valid genetic findings do not validate those inferences. [2,3,25]

A.5 Historical case studies: power and harm

Case 1 — Buck v. Bell, United States, 1927

The US Supreme Court upheld Virginia's compulsory sterilization law in *Buck v. Bell*. Carrie Buck's institutionalization and the hereditary allegations against her were used to authorize irreversible interference with reproduction. Later historical work challenged the evidentiary foundations of the case. The decision is a warning that legal authorization and scientific legitimacy can diverge. [2,4]

The case was not an informed clinical choice to avoid a defined pathogenic variant. It involved state power, contested assessments of disability, and the denial of reproductive autonomy. A modern programme cannot claim ethical legitimacy simply by obtaining a court order or adopting medical language. [4,7]

Case 2 — Nazi compulsory sterilization and racial policy

Germany's 1933 Law for the Prevention of Offspring with Hereditary Diseases authorized compulsory sterilization on diagnostic categories that included conditions interpreted through racial and social ideology. Approximately 400,000 people were forcibly sterilized under Nazi rule. Reproductive coercion existed within a wider system of persecution and escalating violence. [5]

Forced sterilization and the murder of disabled people were distinct policies, although connected by dehumanizing ideas and institutional personnel. The Holocaust also involved goals and mechanisms beyond eugenics; collapsing these events into one undifferentiated programme obscures their history. [5]

Case 3 — Public-health authority and unequal vulnerability

Stern's historical study of California shows how sterilization programmes operated through public-health institutions and intersected with race, gender, immigration, and class. Coercion could be presented as professional expertise or administrative efficiency. The ethical lesson is institutional: safeguards must protect people who have the least practical ability to refuse. [3]

Recurring failure	Present-day safeguard
Social labels treated as biological defects	Validate a specific genotype–condition relationship; challenge biased classifications.
Institutions determine who may reproduce	Protect equal reproductive rights and genuine refusal.
Aggregate benefit overrides individual harm	Require proportionality, independent oversight, and remedies.

B. Scientific basis

B.1 What Mendelian genetics can establish

For some single-gene disorders, a known pathogenic variant and a well-characterized inheritance pattern permit useful risk estimates. If both parents carry the same autosomal-recessive disorder, the simplified Mendelian expectation for each conception is 25% affected genotype, 50% carrier, and 25% without that variant. These are probabilities, not a prediction that every four children will follow this pattern. [10,11,22]

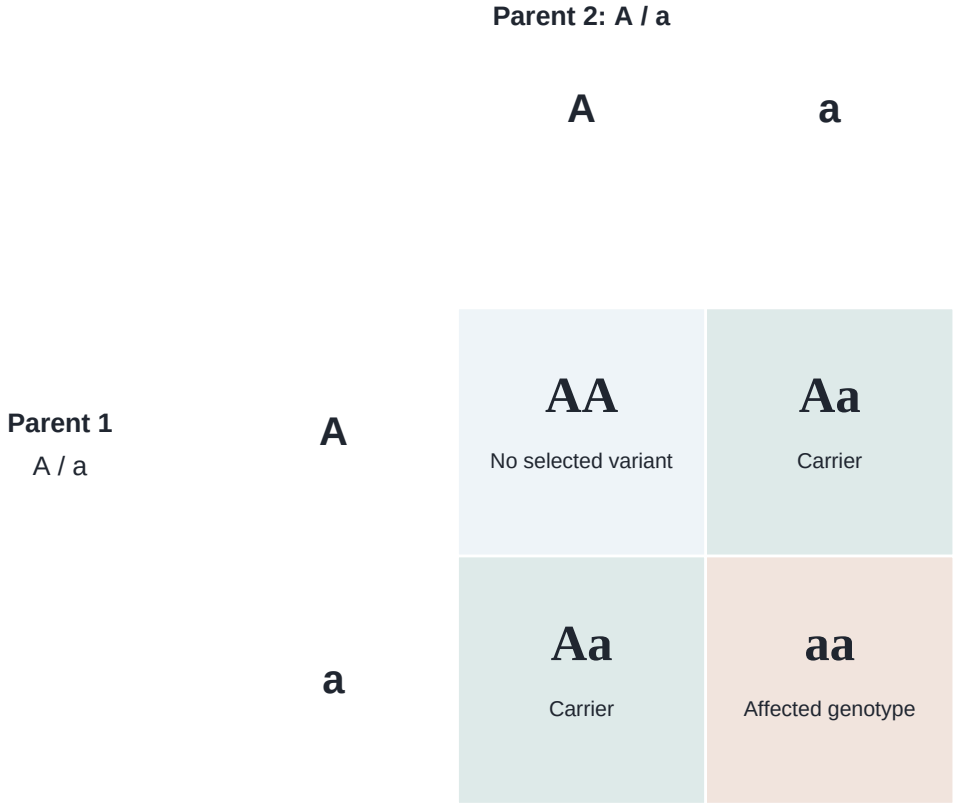


Figure 2. Original Punnett-square illustration for two heterozygous carriers (A/a × A/a). A = reference allele; a = pathogenic allele in this simplified model. This illustration assumes a classic recessive condition and does not model de novo variants, complex alleles, or variable penetrance. [10,11]

B.2 Prediction is conditional

Penetrance is the probability of a phenotype among people with a relevant genotype; **variable expressivity** means its severity or features can differ. Even a strong molecular diagnosis may not precisely predict an individual life. Treatment, age, other variants, and environment can matter. Test error and variant classification also require attention. [10,22]

> What follows—and what does not
> A reliable test may support avoiding transmission of a specified serious disorder. It does not imply that people living with that disorder have less value, that all disabilities are preventable, or that populations can be ranked by genetic “quality.”

B.3 Complex traits and genetic determinism

Most common diseases and behavioural traits are not controlled by one gene. They reflect many variants, each usually contributing a small amount, together with development and environmental conditions. A polygenic score summarizes associations between measured variants and a trait in a particular study setting. It is neither a diagnosis nor a complete biological explanation. [13,15,16]

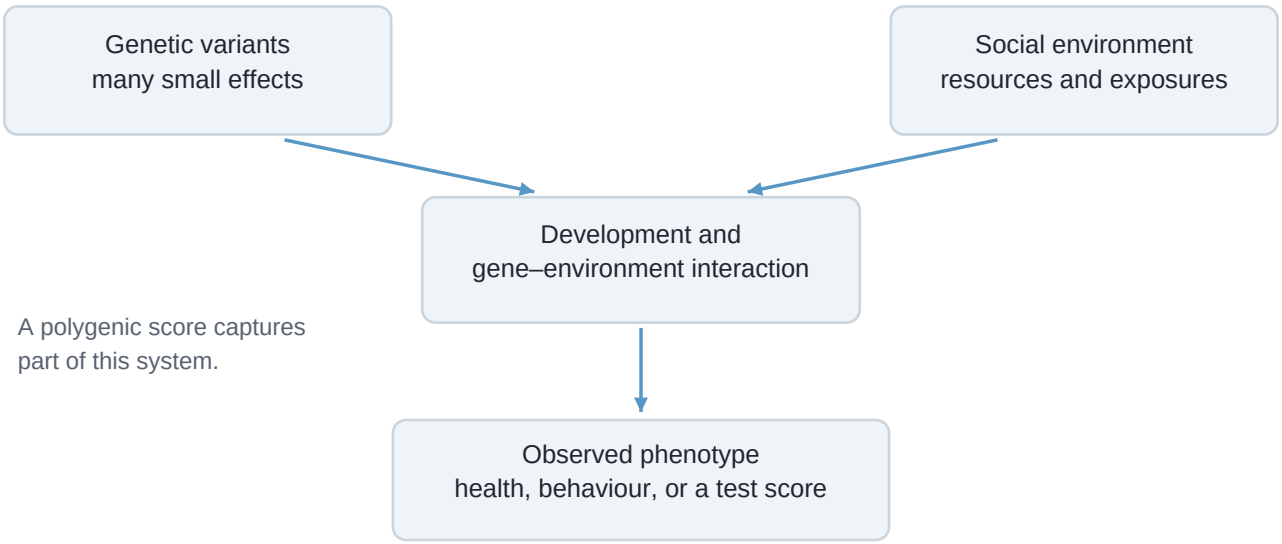


Figure 3. Original conceptual diagram: inherited variants and environment contribute to an observed phenotype. Arrows are explanatory, not estimates of causal effect. Genetic associations may also reflect ancestry structure, family environment, and indirect parental effects. [15,16]

B.4 Heritability does not mean inevitability

Heritability describes the share of observed variation associated with genetic differences in a specified population and environment. It is not the percentage of an individual caused by genes. It does not show that environmental intervention is ineffective, explain differences between social groups, or determine which traits deserve selection. [15,16,24]

B.5 Pleiotropy, ancestry, and family effects

Pleiotropy means a variant can influence more than one trait, creating potential trade-offs. Scores developed largely in one ancestry group may perform differently in another. This is a problem of evidence and calibration, not a ranking of ancestry groups. In addition, parental genotypes can shape a child's environment: associations measured between unrelated people may not transfer unchanged to comparisons among siblings or embryos. [15,16]

Kong and colleagues demonstrated "genetic nurture": parental alleles not transmitted to offspring still predicted aspects of offspring educational attainment through indirect pathways. That result undermines an interpretation in which every genotype–outcome association reflects a direct effect in the future child. [16]

B.6 What polygenic embryo screening can promise

Polygenic embryo screening (often called PGT-P) aims to prioritize embryos by predicted liability to complex conditions or by quantitative traits. Scientific plausibility is not the same as demonstrated clinical benefit. A useful assessment must ask whether the score is valid *within families*, whether the embryo number is realistic, and whether outcomes improve after accounting for IVF burdens. [13–15,21]

Why selection within a family is constrained

Constraint	Consequence for interpretation
Limited embryo numbers	A cycle may yield few or no transferable embryos; large hypothetical pools can overstate practical gains.
Shared parents	Sibling embryos share much of their genetic background. Between-person prediction is not automatically within-family prediction.
Probabilistic outcomes	The embryo with the lowest score can still develop the condition; another embryo may remain unaffected.
Ancestry and calibration	Risk estimates depend on the training data, clinical context, and population baseline.
Multiple selection goals	Prioritizing several risks or traits can reduce gains for each and introduce trade-offs.

Relative benefit versus absolute benefit

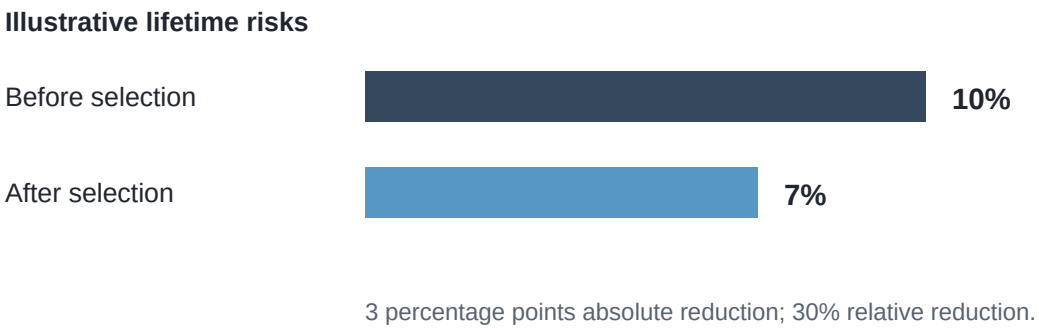


Figure 4. Arithmetic illustration only—not a clinical estimate. A reduction from 10% to 7% is a 30% relative reduction but a 3-percentage-point absolute reduction. Neither number describes the probability of a live birth after an IVF cycle.

Karavani et al. found limited and uncertain gains for polygenic trait selection under the models and data studied. Lencz et al. showed that modelled disease-risk reduction depends strongly on the selection strategy. Together these studies support careful analysis, not a blanket claim of either zero possible benefit or proven routine clinical effectiveness. [13,14]

B.7 Modern reproductive technologies

Genetic counselling should explain the condition, uncertainty, alternatives, and the option to decline testing. “Screening,” “diagnosis,” “selection,” and “editing” describe different actions and should not be used interchangeably. [10,11,22]

Technology	What it does	Main limit or ethical question
Carrier screening	Identifies people who may transmit a recessive or X-linked condition.	Does knowledge expand options, or create pressure to avoid particular relationships or births?
Prenatal testing	Assesses an established pregnancy; some tests screen risk, others are diagnostic.	Screen-positive results may need confirmation; decisions require consent and legal context.
PGT-M	Uses IVF and a condition-specific test to identify embryos in relation to a monogenic disorder.	Cost, IVF burden, embryo availability, residual error, and decisions about embryos.
PGT-A / PGT-SR	Tests chromosome number / structural rearrangements, respectively.	These are distinct from PGT-M and do not establish the value of polygenic screening.
PGT-P	Ranks embryos using scores for complex traits or disease liability.	Within-family utility, calibration, long-term outcomes, equity, and commercial claims.
Heritable editing	Changes DNA before reproduction so alterations may be inherited.	Technical uncertainty, intergenerational effects, and exceptionally demanding governance.

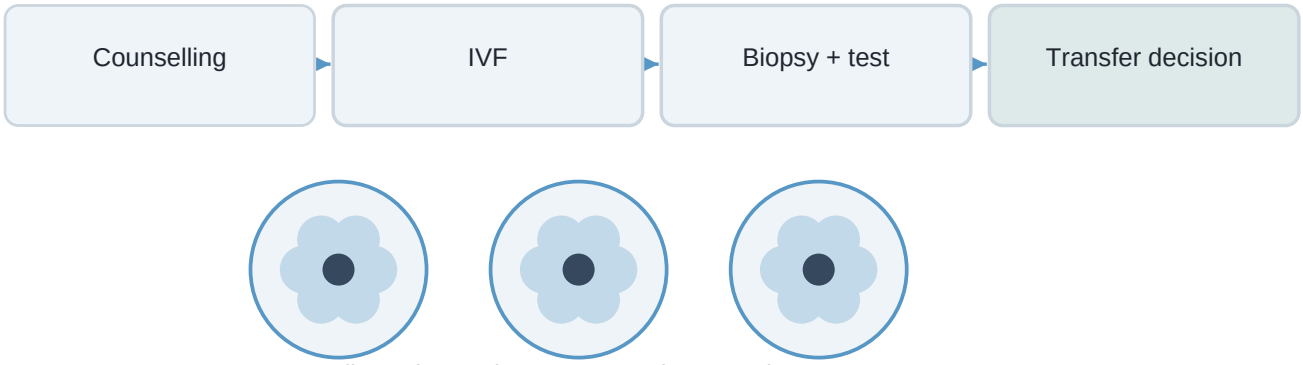


Figure 5. Original schematic of an IVF–PGT-M pathway. Cell illustrations are conceptual, not to scale; actual biopsy, freezing, testing, and transfer schedules differ. Counselling continues throughout. [10,22]

PGT-M selects among embryos; it does not repair the tested DNA variant. A negative result does not guarantee a healthy child. Professional guidance recommends explaining residual diagnostic risk and offering appropriate prenatal confirmation. [10,22]

B.8 Genome editing: a different threshold

Somatic treatment versus heritable alteration

Somatic editing aims to alter cells in the treated person without intentionally transmitting the change to descendants. **Heritable editing** uses altered gametes or embryos in reproduction, potentially affecting future generations. Success in treating an existing patient cannot be assumed to establish the safety or legitimacy of embryo editing. [8,9]

CRISPR-based tools can target DNA, but targeted does not mean perfectly controlled. Relevant concerns include unintended changes away from the target, unwanted changes at the intended site, mosaicism, and uncertain developmental effects. Base and prime editing change the technical possibilities; they do not automatically resolve intergenerational safety or ethical questions. [8,9]

Case study — the He Jiankui affair, 2018

He Jiankui announced births following editing of the *CCR5* gene in embryos, ostensibly to reduce HIV susceptibility. The intervention was widely criticized for inadequate justification, uncertain biological consequences, and failures of ethical governance and informed consent. The embryos were not being treated for an inherited HIV infection; HIV is an infectious disease, and established measures could reduce transmission risk. [20]

The case illustrates why the ability to make an alteration is insufficient. Editing a gene involved in one disease pathway can affect other biological functions, and the altered sequence need not reproduce a well-characterized naturally occurring variant. A child cannot be treated as proof-of-concept for a technology whose benefit is speculative. [9,20]

What governance documents actually support

The 2020 international commission on heritable human genome editing concluded that criteria for precise, reliable changes without undesired alterations had not been met for establishing a pregnancy. It described a tightly limited translational pathway if such uses were ever permitted, rather than endorsing routine reproduction with edited embryos. [9]

WHO’s 2021 recommendations emphasize governance, registries, international coordination, transparency, and responses to unsafe or unregistered activity. They are not a worldwide clinical authorization. National law and scientific guidance differ and can change. [8]

> Scientific and ethical threshold
> A credible benefit, robust safety evidence, meaningful alternatives, public legitimacy, and independent oversight would all be necessary to consider a future use. Better technical accuracy alone would not establish permission.

C. Findings of research and their limits

Different types of evidence answer different questions. A simulation estimates an outcome under assumptions; an association study identifies statistical relationships; historical analysis reconstructs institutions and harms. None alone decides a moral question.

Source and design	What it contributes	What it does not establish
Karavani et al., 2019 [13] Modelling and empirical family-based analysis	Shows limited and variable predicted gains from selecting embryos for quantitative polygenic traits in the settings studied.	A guaranteed change in a child's outcome, or a universal estimate for every future score and IVF setting.
Lencz et al., 2021 [14] Statistical modelling and data-based analysis	Predicts modest reduction from high-risk exclusion and potentially substantial relative reduction from lowest-risk prioritization, given sufficient viable embryos.	Prospectively demonstrated lifetime benefit to children born after a clinical screening programme.
Martin et al., 2019 [15] Analysis of polygenic-score transferability	Explains how uneven ancestry representation can create unequal predictive performance and worsen disparities.	That ancestry is a hierarchy, or that a score is calibrated for every patient.
Kong et al., 2018 [16] Parental transmitted / non-transmitted allele analysis	Shows that parental genotype can influence offspring outcomes through indirect environmental pathways.	That genetic associations are meaningless, or that a score's between-family association is wholly direct.
Stern, 2005 [3] Historical analysis of public-health records	Documents institutional and social inequalities in California's sterilization history.	A clinical estimate of the benefit or harm of an individual modern genetic test.

C.1 How to judge a new claim

Ask what was measured, in whom, with what comparator and uncertainty. For a reproductive claim, specify the denominator: per embryo, transfer, cycle, pregnancy, live birth, or lifetime. Separate analytic validity (does the test measure correctly?), clinical validity (does it predict the condition?), and clinical utility (does using it improve a relevant outcome?). A convincing laboratory result does not supply all three. [10,13–15,22]

Lencz et al. [14] was read in full via eLife's official XML archive. Its empirical check simulated offspring from real genomes; it did not follow children born after screening. The author repositories [23,26] and the paper's bibliography also corroborate the modelling context and key citations. Most other full texts were not directly retrievable.

C.2 Clinical case studies and comparative lessons

Case 4 — Tay–Sachs carrier screening

Tay–Sachs disease is a severe recessive disorder associated with pathogenic variants in *HEXA*. Community carrier screening and prenatal diagnosis became an influential model of reproductive genetics. Kaback and colleagues documented the international experience from 1970 to 1993. It demonstrates how identifying a well-defined inherited risk can inform reproductive decisions. [12]

A reduction in affected births is not, by itself, a complete measure of programme quality. The ethical assessment also asks whether participation is voluntary, counselling is understandable, carriers face stigma, and families receive support regardless of their decisions. A community-based programme can still generate pressure around marriage or reproduction. [11,18]

Case 5 — PGT-M for a serious single-gene condition

Consider a couple with a known familial variant causing a serious, highly penetrant disorder. PGT-M may allow them to attempt a genetically related pregnancy using an embryo not predicted to be affected by that disorder. Scientific justification depends on the specific variant, test design, inheritance, and available embryos; ethical justification depends on burdens, alternatives, and the couple's values. [10,22]

Alternatives may include conceiving without testing, prenatal diagnosis, donor gametes, adoption, or deciding not to have children. Not every option is feasible or acceptable to every family. Counselling should explain these choices without assuming that one reproductive path is a duty. [10,11,22]

Question	Voluntary clinical risk reduction	Coercive population programme
Who sets the goal?	Informed individuals or families, with professional support.	Authorities define desirable people or population targets.
What happens on refusal?	Care, rights, and support remain available.	Penalties, exclusion, or loss of reproductive freedom.
How is success measured?	Accurate information, valued options, proportionate benefit.	Changing the composition of the population, regardless of consent.

These cases show that neither a historical analogy nor a claim of personal choice settles the issue. Examine the actual technology, institutional pressures, and consequences. Responsible clinical genetics must remain compatible with equal respect for people already living with the conditions being tested. [7,18,24]

D. Benefits and limitations

D.1 Strongest arguments for some selective choices

A balanced evaluation should consider the strongest case for a practice, rather than treating all reproductive selection as equivalent to forced sterilization. The arguments below support limited permissions, not an unlimited entitlement to any technology or a state mandate. [17,19,24]

Argument	Why it has force	Condition or unresolved objection
Preventing serious suffering	A family may reasonably wish to lower a well-characterized risk of a severe, life-limiting condition.	Severity, penetrance, available treatment, and IVF burdens must be assessed for the particular condition.
Reproductive autonomy	People have strong interests in making informed decisions about forming a family.	Freedom requires practical ability to refuse; marketing, insurers, and social pressure can constrain it.
Expanding options	Counselling and reliable tests may offer choices beyond accepting a known risk or forgoing a genetic relationship.	Information can also create anxiety; access and support must accompany testing.
Procreative beneficence	Savulescu argues that parents have moral reasons to select the child expected to have the best life among possible children. [17]	“Best life” is contested; forecasts are uncertain and can import ableist or status-based assumptions.
Responsible future prevention	A future intervention might be justifiable if robust benefit and safety are established under legitimate governance.	This is a conditional argument, not evidence that heritable editing is ready or that enhancement is justified.

D.2 The strongest limited conclusion

There can be good reasons to permit voluntary, well-supported selection against a serious inherited condition. The reasons are strongest when evidence is reliable, the expected burden is substantial, less burdensome options are unsuitable, and families receive balanced support. Permission need not imply a duty to select, and declining testing need not be morally blamed. [10,11,18,24]

> The scope of the argument
> A benefit to a family does not justify reproductive quotas, discrimination against carriers, or promises to optimize intelligence, status, or social worth. Those claims require separate evidence and ethical defence.

D.3 Strongest objections and practical limits

1. Disability equality and the expressive concern

Disability scholars argue that routinized selection can communicate that certain lives are less welcome. A person's experienced quality of life may differ from an outsider's clinical forecast, and disadvantage may arise partly from inaccessible environments. This concern does not automatically prohibit every individual choice, but it demands participation by people with lived experience and sustained support for disabled people. [7,18]

2. Choice can become an expectation

A nominally optional test can become difficult to refuse if clinicians present it as routine, insurers condition support on prevention, or parents are blamed for an affected birth. Families need a meaningful right not to know, not to test, and not to select, subject to carefully explained clinical circumstances. Consent is a continuing process, not merely a signed form. [11,18]

3. Inequality and commercialization

Access to IVF and sophisticated testing is unequal. If social advantage influences access to selection and also shapes the outcomes being selected for, a market can reinforce disadvantage while presenting it as biological improvement. Claims about "healthiest" or "best" embryos can obscure uncertainty and promote unrealistic expectations. [15,19,21]

4. Biological uncertainty and competing harms

Complex traits involve pleiotropy, uncertain forecasts, and changing environments. Repeated IVF cycles can impose physical, financial, and emotional burdens. Even useful disease-risk information may have limited practical value when very few embryos are available. Evidence must compare the whole pathway with realistic alternatives, not just compare scores. [10,13,14,22]

5. The future child and the non-identity problem

Selecting one embryo rather than another often determines *which person* will exist. It is therefore philosophically difficult to describe selection as a benefit to the same child who would otherwise have been born with a condition. Ethical evaluation can instead examine foreseeable suffering, parental responsibilities, rights, and fairness without assuming that a disabled life is not worth living. [18,24]

> Limits of "balance"
> Scientific disagreement about a test deserves careful analysis. Forced sterilization and racial hierarchy do not become ethically credible simply because a report presents arguments on both sides. Equal dignity and reproductive rights constrain the debate. [6,7]

E. Ethical implications

E.1 A framework beyond technical accuracy

An ethical judgement needs to combine reproductive freedom with obligations to protect people from harm, discrimination, and unequal power. Human-rights instruments supply a floor of equal dignity and reproductive equality; they do not mechanically resolve every individual case. [6,7]

Principle	What it requires in practice
Autonomy	Understandable counselling; disclosure of uncertainty and alternatives; voluntary consent; a meaningful option to refuse.
Benefit and proportionality	A credible, condition-specific benefit sufficient to justify testing, IVF, uncertainty, and other burdens.
Non-discrimination	No ranking of human worth; no loss of care or rights because of disability, genotype, ancestry, or reproductive choice.
Justice	Fair access, attention to opportunity costs, protection from discriminatory use, and support for families who decline selection.
Privacy	Proportionate data collection, secure handling of familial genetic information, and clear policies on secondary use.
Accountability	Independent review, transparent claims, public participation, outcome monitoring, and routes to challenge abuse.

E.2 Important disagreements remain

People disagree about the moral status of embryos and the permissibility of embryo testing, storage, and non-transfer. Those views affect assessments of PGT-M even when its prediction is reliable. There are also reasonable disagreements about what counts as a serious condition, how adult-onset risks should be treated, and how to balance parental choice with future autonomy. [19,24]

These disputes require case-specific deliberation and applicable law. They should not be hidden behind words such as “health,” “normality,” or “improvement.” Equally, respect for pluralism does not permit threats, forced sterilization, or denial of reproductive rights to disabled people. [6,7]

E.3 Institutional legitimacy

Professionals should separate counselling from sales incentives, explain absolute as well as relative risks, and include disability perspectives. Regulators should scrutinize claims and provide accessible complaint routes. For heritable editing, consequences extend beyond one clinical encounter; international cooperation and public accountability are especially important. [8,11,18,19]

E.4 Decision framework for a proposed use

The following flowchart is an original synthesis of the report's ethical analysis. It is a tool for structured deliberation, not a validated clinical guideline or a substitute for applicable law. Passing every stage supports consideration; it does not guarantee approval. [6–11,18,19,22]

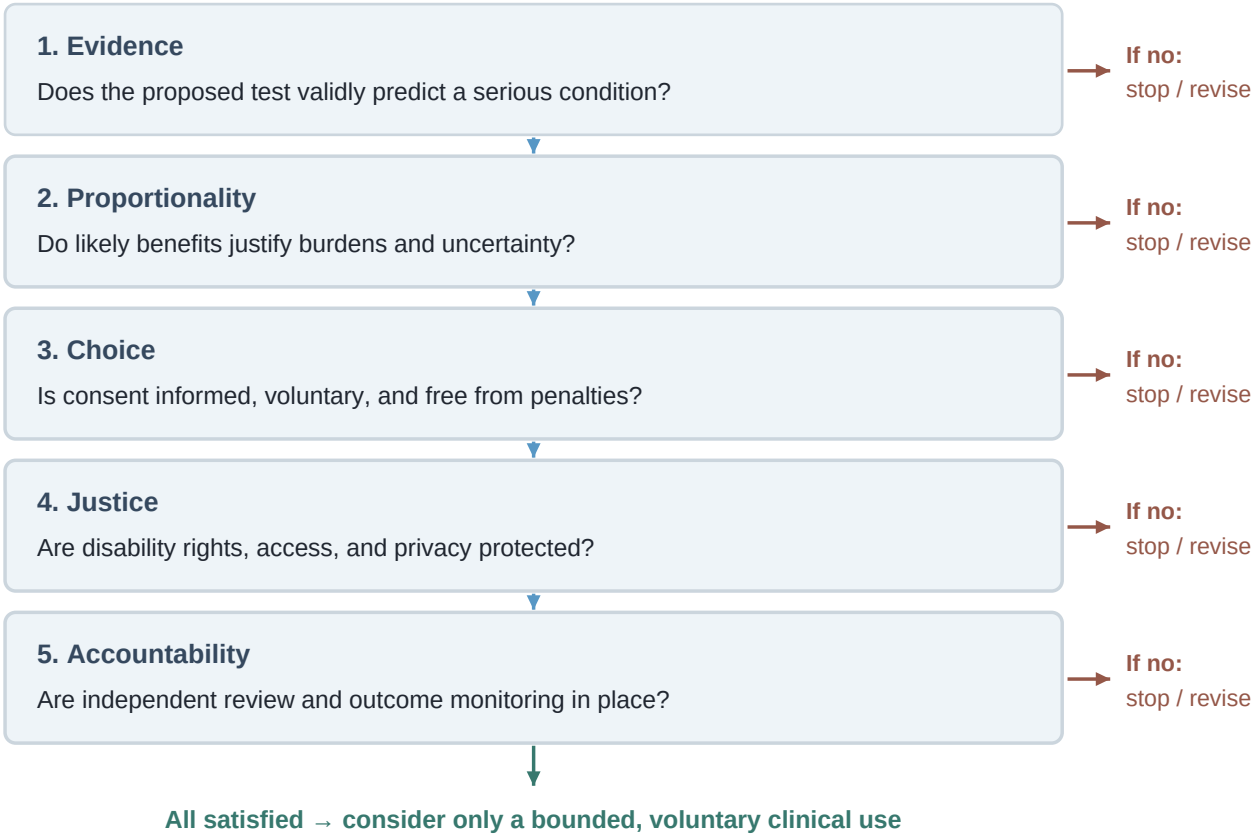


Figure 6. Five linked requirements for evaluating reproductive genetic selection. A failure at any step calls for rejection, revision, or further evidence. Heritable editing requires additional intergenerational and technical scrutiny.

Worked comparison

Proposal	Assessment using the framework
Mandatory sterilization of people with an inherited condition	Fails choice, equal rights, and proportionality regardless of test accuracy. Reject.
Voluntary PGT-M for a severe, well-characterized disorder	Potentially defensible after individualized counselling, reliable testing, and assessment of burdens and alternatives.
Marketing polygenic ranking as a guaranteed "best child" service	Fails evidence and honest consent. Probabilistic scores cannot establish a uniquely best future life.

A crucial counterfactual is whether the same family would still receive care, financial support, and respect after refusing the test or choosing to continue an affected pregnancy. If not, the programme's claim of voluntariness is weakened. [7,11,18]

F. Future perspectives and research priorities

F.1 What better evidence should establish

Future studies of embryo selection should distinguish modelled gains from clinically meaningful outcomes. Priorities include within-family prediction, ancestry-appropriate calibration, realistic embryo availability, cumulative outcomes across an IVF pathway, and prospective evaluation where ethically appropriate. Long-term follow-up needs independent oversight and privacy protection. [13–15,21]

Researchers should specify the comparator: standard IVF, PGT-M, no selection, or another reproductive pathway. Reports should separate per-transfer outcomes from cumulative live-birth outcomes and longer-term health. Null results, limitations, and uncertainty should be published alongside promising findings. [10,14,22]

F.2 Technology does not remove value choices

Improved sequencing, better scores, or more precise editing could change what is technically possible. They cannot decide which traits matter, how much risk is acceptable, or who should control reproduction. Selection for an adult-onset risk raises different questions from selection against a severe childhood disorder; enhancement claims raise further problems of social norms and competitive pressure. [9,19,24]

F.3 Governance and participation

Policy should include people affected by genetic conditions, disability communities, clinicians, ethicists, and the wider public. Oversight must examine commercial incentives and hidden coercion as well as technical error. Global coordination matters because permissive jurisdictions and cross-border services can undermine local safeguards. [8,18,19]

Priority	A concrete research or policy question
Clinical utility	Does using the test improve outcomes that families value, compared with a realistic alternative?
Equity	Who benefits, who pays, and whose ancestry is underrepresented in validation?
Voluntariness	Can people refuse without financial penalties, stigma, or loss of support?
Disability inclusion	How do affected people describe quality of life, support needs, and the meaning of prevention?
Long-term accountability	Who monitors outcomes and responds when predictions or commercial claims fail?

A constructive direction
Invest in accurate counselling and proportionate prevention while improving treatment, accessibility, and social support. Reducing the burden of disease and respecting existing lives are compatible obligations.

G. Conclusion: can it ever be justified?

G.1 Scientifically: only a specific, testable claim

Some reproductive interventions can meaningfully reduce the probability of a particular inherited condition. The strongest examples involve well-characterized monogenic disorders and reliable testing. That conclusion is bounded by inheritance, penetrance, test limitations, embryo availability, and the burdens of the reproductive pathway. It is not evidence that human beings can be arranged on a single genetic scale of quality. [10,11,22]

Complex-trait selection is more uncertain. Polygenic models can estimate potential benefits, but estimates depend on scores, ancestry, family context, selection strategy, and available embryos. Heritability is not destiny, population-level association is not automatic within-family utility, and a risk reduction is not a guarantee of health. Heritable editing adds technical and intergenerational concerns. [8,9,13–16]

G.2 Ethically: permission is narrower than a programme

A voluntary reproductive choice to reduce a serious inherited risk may be justified when it rests on credible evidence, informed consent, proportionate burdens, genuine alternatives, and protections for equality. It should remain a choice rather than a duty, and its availability must not reduce support for people who live with the condition or decline selection. [7,11,18,24]

Coercive eugenics fails this test. Forced sterilization, racial hierarchy, reproductive penalties, and population “improvement” targets subordinate persons to institutional goals. Better genetic prediction would not cure that ethical defect. The historical record explains why safeguards must address power and discrimination, not just laboratory accuracy. [2–7,25]

G.3 Final judgement

> A qualified answer
> Yes: some carefully bounded forms of voluntary selection to reduce serious inherited disease risk can be scientifically and ethically defensible. No: that does not justify coercive eugenics or a general project to breed “better” people. The defensible object is an evidence-based reproductive option that respects equal human worth—not a hierarchy of lives.

Calling every use of reproductive genetics “eugenics” can hide important distinctions; calling every consumer choice “autonomous” can hide social pressure. The responsible position is to examine each proposed practice, retain the right to refuse, and demand that scientific promises remain accountable to the people whose lives they shape. [11,18,19,24]

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